HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 8: Sample and Hold

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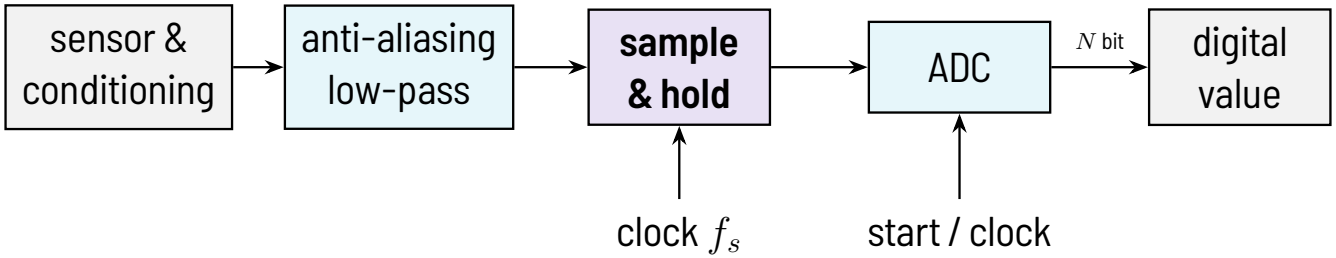
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8 Sample and Hold

8.1 Why Sample and Hold?

Position in the signal chain



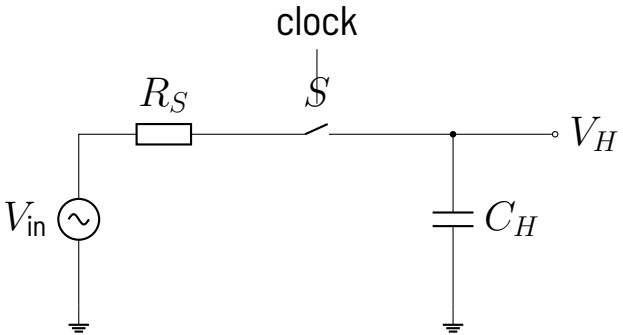
- the S&H stage **defines the sampling instant** $t_k = k\,T_s$, $T_s = 1/f_s$
- it keeps the voltage **constant while the ADC converts**
- capstone: $f_s = 5$ Hz, 3-bit flash converter, full scale 4 V

What goes wrong without S&H

- the input changes during the conversion time t_{conv}
- the error must stay below $\frac{1}{2}$ LSB: $\left|\frac{dV}{dt}\right|_{\text{max}} t_{\text{conv}} < \frac{1}{2}\,\text{LSB}$
- sine $V(t) = \hat{V}\sin(2\pi f t)$: maximum slope $2\pi f \hat{V}$
- example: 12-bit ADC, 4 V full scale (LSB $= 0.98$ mV), $t_{\text{conv}} = 10$ µs, $\hat{V} = 2$ V
  - $f < \frac{\text{LSB}/2}{2\pi \hat{V} t_{\text{conv}}} = 3.9$ Hz without S&H
  - with S&H only the much shorter **aperture time** counts (nanoseconds)
- a **flash** converter decides in one step, but the digital outputs still have to be read and displayed at a defined instant

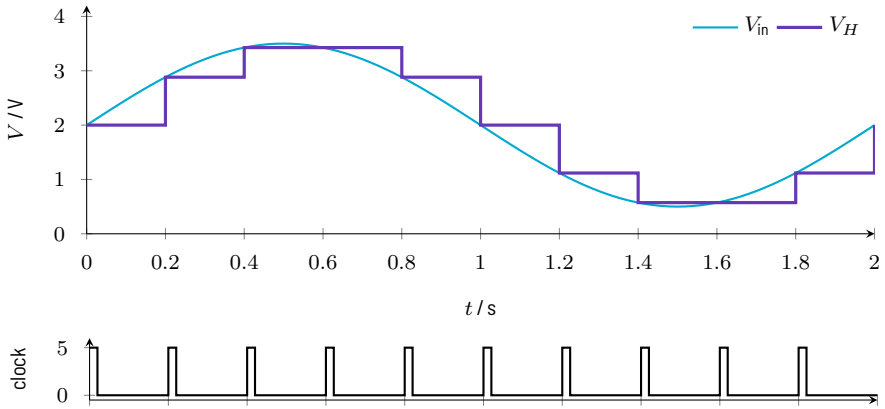
8.2 The Track-and-Hold Principle

Switch and hold capacitor



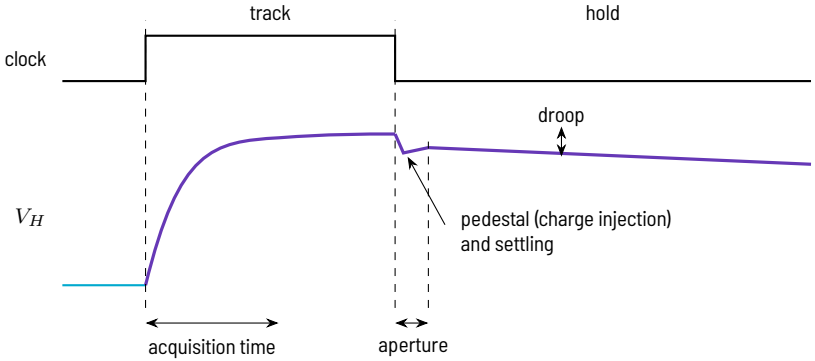
- clock high: switch closed, C_H follows V_{in} (**track**)
- clock low: switch open, C_H keeps its charge (**hold**)
- capstone clock: 5 Hz, duty cycle $10\,\%$: track 20 ms, hold 180 ms

Timing diagram



- the output is a **staircase**: one step per sampling period
- the value of each step is V_{in} at the **end** of the track phase

Terms



- **acquisition time**: track time needed until V_H is within the error band (e.g. $\frac{1}{2}$ LSB)
- **aperture time**: delay between the hold command and the real opening of the switch; its variation is the **aperture jitter**
- **pedestal**: step at the transition to hold; **droop**: slow change during hold

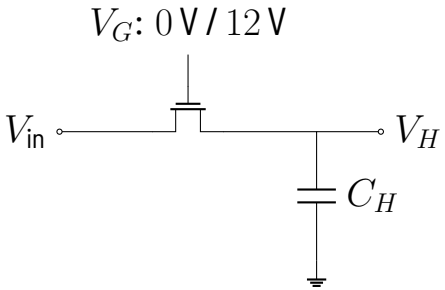
Falstad: basic S&H

- $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\ \sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$, $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$
- clock rail: square wave 5 Hz, 0 V … 5 V, duty cycle 0.1
- held values: 2.094 V, 2.956 V, 3.452 V, 3.394 V, 2.804 V
- the scope shows V_{in}, V_H (label `Vhold`) and the clock

▶ Falstad: `examples/samplehold/sh_basic.txt`

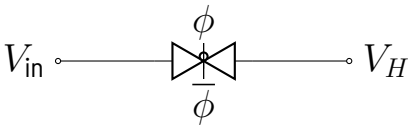
8.3 Analog Switches

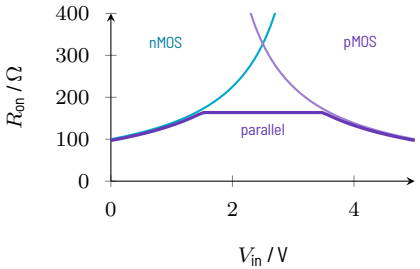
The MOSFET as a switch



- n-channel MOSFET: conducts if $V_{GS} = V_G - V_{\text{in}} > V_{th}$
- on-resistance $R_{\text{on}} \approx \frac{1}{k\,(V_{GS} - V_{th})}$: depends on V_{in}!
- near $V_{\text{in}} \approx V_G - V_{th}$ the switch does not close any more
- off: leakage currents in the pA ...nA range

CMOS transmission gate





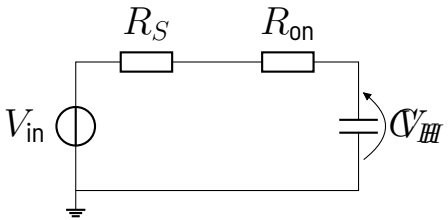
- nMOS and pMOS in parallel, driven with complementary signals
- nMOS conducts well near 0 V, pMOS near V_{DD}: together almost constant R_{on}
- integrated as CD4066, 74HC4066 (4 switches, $R_{\text{on}} \approx 50\,\Omega \ldots 300\,\Omega$), DG4xx

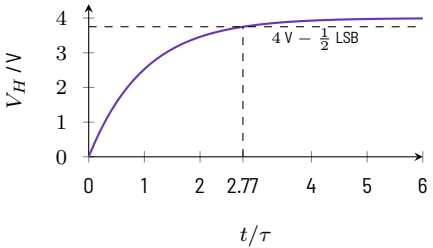
The analog switch in Falstad

- element **Analog Switch** (models a FET switch / transmission gate)
  - closed if the control input is above 2.5 V
  - $R_{\text{on}} = 20\,\Omega$, $R_{\text{off}} = 10$ GΩ (defaults, edit with a double click)
- no charge injection, no leakage: add them as extra elements if you want to see their effect
- in our circuits: $R_S = 1$ kΩ in series
  - limits the current from the op-amp output into the empty capacitor
  - isolates the op-amp from the large capacitive load (stability)
  - real lab switches have R_{on} of some $100\,\Omega$ anyway

8.4 Designing the Hold Capacitor

Acquisition: charging in the track window





- $V_H(t) = V_{\text{in}}\,(1 - e^{-t/\tau})$ with $\tau = (R_S + R_{\text{on}})\,C_H$
- error after time t: $V_{\text{in}}\,e^{-t/\tau}$; worst case: step over the full scale V_{FS}

Settling to $\frac{1}{2}$ LSB

- requirement: $V_{\text{FS}}\, e^{-t/\tau} \leq \frac{1}{2}\,\text{LSB} = \frac{V_{\text{FS}}}{2^{N+1}}$

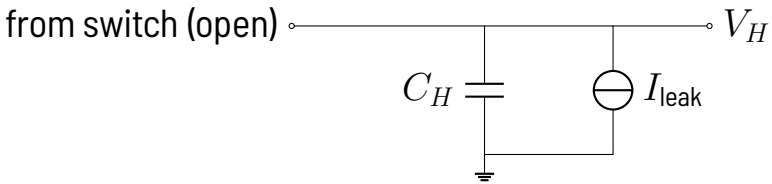
Acquisition time for N bit

$$t_{\text{acq}} = \tau\, \ln\left(2^{N+1}\right) = (N+1)\, \ln 2 \cdot \tau \approx 0.69\,(N+1)\,\tau$$

- $N = 3$: $t_{\text{acq}} = 2.77\,\tau$; $N = 8$: $6.2\,\tau$; $N = 12$: $9.0\,\tau$; $N = 16$: $11.8\,\tau$
- design rule: $t_{\text{acq}} \leq t_{\text{track}} \quad \Rightarrow \quad C_H \leq \dfrac{t_{\text{track}}}{(R_S + R_{\text{on}})\, \ln 2^{N+1}}$
- example: $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$: $\tau = 1.02\,\text{ms}$, $t_{\text{acq}} = 2.83\,\text{ms} \ll 20\,\text{ms}$

▶ Falstad: `examples/samplehold/sh_acquisition.txt`

Droop: losing charge in the hold phase



- leakage currents: switch (off), capacitor dielectric, input bias current of the buffer, board surface

Droop

$$\frac{dV_H}{dt} = \frac{I_{\text{leak}}}{C_H} \quad \Rightarrow \quad \Delta V_{\text{droop}} = \frac{I_{\text{leak}}\, t_{\text{hold}}}{C_H} \quad \Rightarrow \quad C_H \geq \frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}}$$

- t_{hold}: until the conversion is finished; worst case the whole period $T_s - t_{\text{track}}$

Too small, right, too large

- Falstad: $V_{\text{in}} = 3\,\text{V}$, $R_S = 1\,\text{k}\Omega$, leakage $I_{\text{leak}} = 50\,\text{nA}$ (exaggerated)

	$C_H = 10\,\text{nF}$	$C_H = 1\,\mu\text{F}$	$C_H = 100\,\mu\text{F}$
τ	$10\,\mu\text{s}$	$1.02\,\text{ms}$	$102\,\text{ms}$
charged in $20\,\text{ms}$?	yes	yes	only $18\,\%$ per sample
droop in $180\,\text{ms}$	$0.9\,\text{V}$	$9\,\text{mV}$	$0.09\,\text{mV}$
Falstad	$3.00\,\text{V} \to 2.10\,\text{V}$	$3.00\,\text{V} \to 2.99\,\text{V}$	$0.534\,\text{V}, \ldots, 1.876\,\text{V}$ (5th)

▶ Falstad: `examples/samplehold/sh_capsize.txt`

Choosing C_H: the window

- upper limit from the acquisition, lower limit from the droop:

Hold capacitor

$$\frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}} \leq C_H \leq \frac{t_{\text{track}}}{(R_S + R_{\text{on}}) \ln 2^{N+1}}$$

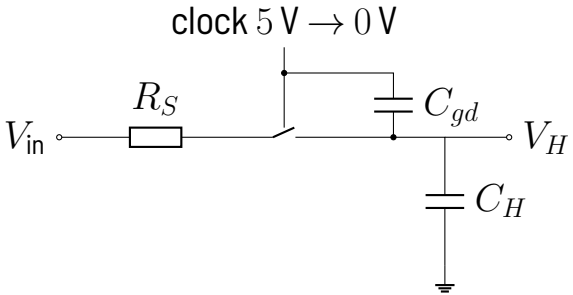
- capstone values: $I_{\text{leak}} = 10\,\text{nA}$, $\Delta V_{\text{max}} = \frac{1}{10}\,\text{LSB} = 50\,\text{mV}$, $t_{\text{hold}} = 180\,\text{ms}$, $t_{\text{track}} = 20\,\text{ms}$, $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$
  - $36\,\text{nF} \leq C_H \leq 7.1\,\mu\text{F} \quad \Rightarrow \quad C_H = 1\,\mu\text{F}$ (geometric middle, film capacitor)
- choose the capacitor type: polypropylene or polystyrene film (low leakage, low dielectric absorption), C0G ceramic for small values; no electrolytic capacitors

Design card: Sample and Hold

- structure: input follower $\rightarrow$ R_S + switch $\rightarrow$ C_H to ground $\rightarrow$ output follower
- clock: f_s, short duty cycle; track $t_{\text{track}} = D\,T_s$ (capstone: 5 Hz, $10\,\%$: 20 ms)
- charging: $\tau = (R_S + R_{\text{on}})\,C_H$, settled to $\frac{1}{2}$ LSB after $\tau \ln 2^{N+1}$ ($N = 3$: $2.77\,\tau$) $\leq t_{\text{track}}$
- droop: $\Delta V = I_{\text{leak}}\,t_{\text{hold}}/C_H \leq$ fraction of an LSB
- pedestal: $\Delta V = C_{gd}\,\Delta V_{\text{clk}}/(C_H + C_{gd})$; larger C_H helps
- capstone choice: $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu$F film: $t_{\text{acq}} = 2.8$ ms, droop 1.8 mV at 10 nA

8.5 Errors of a Real Sample and Hold

Charge injection (pedestal)



- the gate of the switch is coupled to the hold node through C_{gd} (and the channel charge)
- when the switch opens, the clock step ΔV_{clk} is divided by C_{gd} and C_H:

$$\Delta V_{\text{ped}} = -\Delta V_{\text{clk}} \frac{C_{gd}}{C_{gd} + C_H} \approx -\frac{Q_{\text{inj}}}{C_H}$$

- $C_{gd} = 10\,\text{pF}$, $5\,\text{V}$: $C_H = 1\,\text{nF}$: $-49.5\,\text{mV}$; $C_H = 100\,\text{nF}$: $-0.5\,\text{mV}$

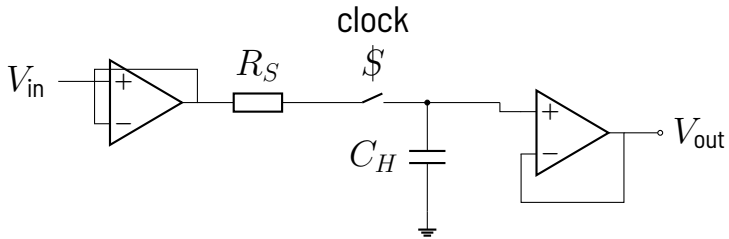
▶ Falstad: `examples/samplehold/sh_pedestal.txt`

Feedthrough, aperture and jitter

- **feedthrough**: the open switch has a capacitance C_{ds}; the input still reaches C_H: $\Delta V_H = \Delta V_{\text{in}} \frac{C_{\text{ds}}}{C_{\text{ds}}+C_H}$
- **aperture delay** t_a: the switch opens a little after the clock edge; constant, only shifts the sampling instant
- **aperture jitter** t_j: random variation of the sampling instant
  - error $\Delta V = \frac{dV}{dt}\, t_j$; sine: $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j$
  - requirement $\Delta V_{\text{max}} < \frac{1}{2}$ LSB: $t_j < \frac{1}{2\pi f\, 2^{N+1}} \cdot \frac{V_{\text{FS}}}{\hat{V}}$
  - example: 12 bit, full-scale sine of 1 MHz: $t_j < 19$ ps
- **dielectric absorption**: held voltage creeps back towards the previous value

8.6 The Buffered Sample and Hold

Follower – switch – capacitor – follower



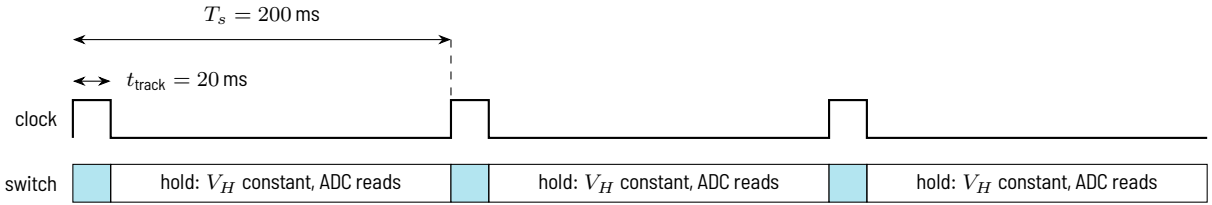
- **input buffer**: the charging current comes from the op-amp, not from the signal source
- **output buffer**: draws (almost) no current from C_H; only its input bias current adds to the leakage

Why both buffers are needed

- without buffers: source $R_i = 10\,\text{k}\Omega$, load $R_L = 100\,\text{k}\Omega$ (e.g. ADC input), $V_\text{in} = 3\,\text{V}$
  - track: $\tau = (R_i \parallel R_L)\, C_H = 9.1\,\text{ms}$, final value $3\,\text{V} \cdot \frac{100}{110} = 2.73\,\text{V}$
  - after $20\,\text{ms}$: $2.425\,\text{V}$
  - hold: discharge through R_L, $\tau = 100\,\text{ms}$: $0.401\,\text{V}$ after $180\,\text{ms}$
- with buffers: $3.000\,\text{V}$ during the whole period

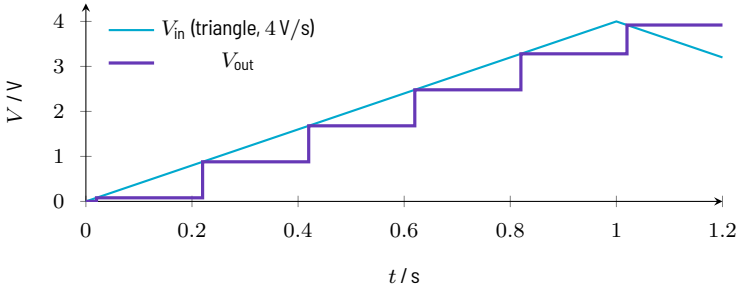
▶ Falstad: `examples/samplehold/sh_buffered.txt`

The sampling clock



- square wave with a short duty cycle $D = t_{\text{track}}/T_s$
- Falstad: rail with waveform **square**, 5 Hz, amplitude 2.5 V, offset 2.5 V, duty cycle 0.1
- in hardware: 555 timer or microcontroller output; 0/5 V logic level drives the switch
- the ADC result is read in the hold phase (e.g. in the middle or at the end)

Sampling a ramp



- buffered S&H, triangle 0 V … 4 V at 0.5 Hz: steps of $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8$ V
- Falstad: 0.08 V, 0.88 V, 1.68 V, 2.48 V, 3.28 V

▶ Falstad: `examples/samplehold/sh_staircase.txt`

Practical hints

- op-amps: rail-to-rail output for the full range $0\,\mathrm{V} \ldots 4\,\mathrm{V}$ on a single $5\,\mathrm{V}$ supply, FET input for the output buffer
- guard ring around the hold node on the PCB: surface leakage
- integrated S&H amplifiers (LF398, AD783) contain both buffers and the switch; only C_H is external
- in most modern ADCs (SAR, sigma-delta) the S&H is built in: a switched capacitor array at the input
  - the driving op-amp must recharge this capacitor in the acquisition time: the same calculation as above

8.7 Exercises

Exercise 8.1: Choosing the hold capacitor for $f_s = 5\,\text{Hz}$

A buffered S&H stage samples at $f_s = 5$ Hz with a clock duty cycle of $10\,\%$. The charging path consists of $R_S = 1\,\text{k}\Omega$ and a switch with $R_{\text{on}} = 20\,\Omega$. The total leakage current at the hold node (switch, capacitor, buffer bias) is at most $I_{\text{leak}} = 10$ nA. The stage drives a 3-bit converter with $V_{\text{FS}} = 4$ V.

a) Calculate the track and hold times.

b) The capacitor must settle to $\frac{1}{2}$ LSB for a full-scale step within the track phase. Find the upper limit for C_H.

c) The droop during the hold phase must stay below $\frac{1}{10}$ LSB. Find the lower limit for C_H.

d) Choose C_H and calculate acquisition time and droop for your choice.

e) Check the charging curve in Falstad.

Solution 8.1

a) $T_s = 1/f_s = 200\,\text{ms}$, $t_{\text{track}} = 0.1 \cdot T_s = 20\,\text{ms}$, $t_{\text{hold}} = 180\,\text{ms}$.

b) LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$. Settling to $\frac{1}{2}$ LSB needs $t = \tau \ln 2^4 = 2.77\,\tau \le 20\,\text{ms}$, so $\tau \le 7.21\,\text{ms}$ and $C_H \le 7.21\,\text{ms}/1020\,\Omega = 7.1\,\mu\text{F}$.

c) $\Delta V_{\text{max}} = 50\,\text{mV}$: $C_H \ge I_{\text{leak}}\, t_{\text{hold}}/\Delta V_{\text{max}} = 10\,\text{nA} \cdot 0.18\,\text{s}/50\,\text{mV} = 36\,\text{nF}$.

d) The window is $36\,\text{nF}$ …$7.1\,\mu\text{F}$; the geometric middle is about $0.5\,\mu\text{F}$. We choose $C_H = 1\,\mu\text{F}$ (film capacitor). Then $\tau = 1.02\,\text{ms}$, $t_{\text{acq}} = 2.77\,\tau = 2.83\,\text{ms}$ (seven times faster than required) and $\Delta V_{\text{droop}} = 10\,\text{nA} \cdot 0.18\,\text{s}/1\,\mu\text{F} = 1.8\,\text{mV}$ (28 times smaller than allowed).

e) ▶ Falstad: `examples/samplehold/sh_acquisition.txt`

With a $4\,\text{V}$ input, `Vhold` shows $2.53\,\text{V}$ after $1.02\,\text{ms}$, $3.75\,\text{V}$ after $2.83\,\text{ms}$ and $4.00\,\text{V}$ at the end of the track phase. (Pause the simulation and use the scope, or reduce the simulation speed.)

Exercise 8.2: Droop and a too large capacitor

The S&H stage of exercise 8.1 has an (exaggerated) leakage current of 50 nA. The input is a constant 3 V.

a) Calculate the droop rate and the voltage at the end of the first hold phase for $C_H = 10$ nF and for $C_H = 1$ µF.

b) With $C_H = 100$ µF, which fraction of the remaining difference is charged in one track window? Calculate the held voltage after the first and after the fifth sample (leakage negligible).

c) Compare with Falstad.

Solution 8.2

a) $C_H = 10\,\text{nF}$: $dV/dt = 50\,\text{nA}/10\,\text{nF} = 5\,\text{V/s}$, droop in $180\,\text{ms}$: $0.9\,\text{V}$, the held voltage falls from $3.00\,\text{V}$ to $2.10\,\text{V}$. $C_H = 1\,\mu\text{F}$: $50\,\text{mV/s}$, droop $9\,\text{mV}$.

b) $\tau = 1.02\,\text{k}\Omega \cdot 100\,\mu\text{F} = 102\,\text{ms}$, fraction $a = 1 - e^{-20/102} = 0.178$. After n samples $V_n = 3\,\text{V}\,(1-(1-a)^n)$: $V_1 = 0.534\,\text{V}$, $V_5 = 3\,\text{V}\,(1-0.822^5) = 1.876\,\text{V}$.

c) ▶ Falstad: `examples/samplehold/sh_capsize.txt`

`V_10n` drops from $3.00\,\text{V}$ to $2.10\,\text{V}$ in every hold phase, `V_1u` loses $9\,\text{mV}$ ($3.00\,\text{V} \rightarrow 2.99\,\text{V}$), `V_100u` climbs in steps: $0.534\,\text{V}$ after the first, $1.876\,\text{V}$ after the fifth sample.

Exercise 8.3: Unbuffered sample and hold

A switch and $C_H = 1$ µF are connected directly between a source ($V_{\text{in}} = 3$ V, internal resistance $R_i = 10$ kΩ) and a load $R_L = 100$ kΩ. Clock: 5 Hz, 10 %. Neglect R_{on}.

a) Draw the equivalent circuit for the track phase and calculate its final value and time constant.
b) Calculate V_H at the end of the first track phase and at the end of the first hold phase.
c) How does the buffered circuit change the result? Check in Falstad.

Solution 8.3

a) Thevenin equivalent seen by C_H: $V_{\text{th}} = 3\,\text{V} \cdot \frac{100}{110} = 2.727\,\text{V}$, $R_{\text{th}} = R_i \parallel R_L = 9.09\,\text{k}\Omega$, $\tau = 9.09\,\text{ms}$.

b) End of track: $V_H = 2.727\,\text{V}\,(1 - e^{-20/9.09}) = 2.425\,\text{V}$. Hold: C_H discharges through R_L with $\tau = 100\,\text{ms}$: $V_H = 2.425\,\text{V} \cdot e^{-1.8} = 0.401\,\text{V}$.

c) The input follower removes the divider and the slow charging, the output follower removes the discharge: $V_{\text{out}} = 3.000\,\text{V}$ throughout.

▶ Falstad: `examples/samplehold/sh_buffered.txt`

`Vout_unbuf` shows $2.425\,\text{V}$ and $0.401\,\text{V}$, `Vout_buf` $3.000\,\text{V}$.

Exercise 8.4: Pedestal error

The gate of the switch has a coupling capacitance $C_{gd} = 10$ pF to the hold node. The clock falls from 5 V to 0 V when the switch opens.

a) Calculate the pedestal for $C_H = 1$ nF and $C_H = 100$ nF.
b) Which C_H is needed for a pedestal below $\frac{1}{2}$ LSB of a 12-bit converter with $V_{\text{FS}} = 4$ V?
c) Is the pedestal a problem for the 3-bit capstone with $C_H = 1$ µF?

Solution 8.4

a) $\Delta V = -5\,\text{V} \cdot \frac{10\,\text{pF}}{1.01\,\text{nF}} = -49.5\,\text{mV}$; with $100\,\text{nF}$: $-0.50\,\text{mV}$.
▶ Falstad: `examples/samplehold/sh_pedestal.txt`: $2\,\text{V}$ input, held values $1.9505\,\text{V}$ and $1.9995\,\text{V}$.

b) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{13} = 0.49\,\text{mV}$: $C_H \geq 5\,\text{V} \cdot 10\,\text{pF}/0.49\,\text{mV} = 0.1\,\mu\text{F}$.

c) $\Delta V = 50\,\mu\text{V}$, i.e. 10^{-4} of an LSB of $0.5\,\text{V}$: no problem.

Exercise 8.5: Ramp input and step size

A buffered S&H stage ($f_s = 5$ Hz) samples a triangle that rises from 0 V to 4 V in 1 s.

a) Calculate the height of the steps of the output staircase.

b) The sampling instants are at the end of the track windows, $t_k = 20\,\text{ms} + k \cdot 200\,\text{ms}$. List the first five held values.

c) What is the maximum slope of the input if consecutive samples may differ by at most one LSB of the 3-bit converter? What does this mean for a sine with an amplitude of 2 V?

Solution 8.5

a) $4\,\mathrm{V/s} \cdot 0.2\,\mathrm{s} = 0.8\,\mathrm{V}$.

b) $V(t_k) = 4\,\mathrm{V/s} \cdot t_k$: $0.08\,\mathrm{V}$, $0.88\,\mathrm{V}$, $1.68\,\mathrm{V}$, $2.48\,\mathrm{V}$, $3.28\,\mathrm{V}$.

▶ Falstad: `examples/samplehold/sh_staircase.txt`: `Vout` shows the same values (4 mV lower, because C_H lags by $\tau \cdot 4\,\mathrm{V/s}$).

c) $0.5\,\mathrm{V}/0.2\,\mathrm{s} = 2.5\,\mathrm{V/s}$. Sine: $2\pi f \cdot 2\,\mathrm{V} \leq 2.5\,\mathrm{V/s}$ gives $f \leq 0.2\,\mathrm{Hz}$. Faster signals are still sampled correctly as long as $f < f_s/2$ (next chapter), but the codes of neighbouring samples jump.

Exercise 8.6: Aperture jitter

The 5 Hz clock of the capstone comes from a microcontroller whose software has a timing uncertainty of $t_j = 1$ ms. The input is a sine with $\hat{V} = 2$ V and $f = 2.5$ Hz (the highest frequency the anti-aliasing filter passes).

a) Calculate the maximum sampling error caused by the jitter.

b) Compare with $\frac{1}{2}$ LSB of the 3-bit converter.

c) Which jitter would a 16-bit converter tolerate for the same signal ($V_{\text{FS}} = 4$ V)?

Solution 8.6

a) $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j = 2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V} \cdot 1\,\text{ms} = 31\,\text{mV}$.

b) $\frac{1}{2}\,\text{LSB} = 250\,\text{mV}$: the jitter error is eight times smaller, harmless.

c) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{17} = 30.5\,\mu\text{V}$: $t_j < 30.5\,\mu\text{V}/(2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V}) = 0.97\,\mu\text{s}$. A software-timed clock is no longer good enough; a hardware timer is needed.